

Vol 6 Chapter 7 — Quantum Statistical Mechanics

Keeper (author pass — deep math/physics revision)

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Chapter 7 — Quantum Statistical Mechanics

Level 1 — one sentence

Quantum statistical mechanics — Bose-Einstein and Fermi-Dirac distributions, blackbody radiation, Bose-Einstein condensation, Fermi gases, ideal-gas heat capacities at low temperature — is the substrate's K-type spectrum populated with Pin(2)-determined particle statistics (bosons for integer-spin K-types, fermions for half-integer), with the discrete substrate spectrum producing all the quantum corrections to classical equipartition.

Level 2 — graduate-physicist precision

7.1 Indistinguishable particles in stat mech

Quantum statistical mechanics applies to identical indistinguishable particles. The grand canonical partition function for a system of identical particles in single-particle states $\{|k\rangle\}$ with energies ϵ_k :

$$\Xi = \prod_k \Xi_k$$

where the per-mode grand partition function depends on the particle statistics:

- Bosons (any occupation number $n_k = 0, 1, 2, \dots$): $\Xi_k^B = \sum_{n=0}^{\infty} e^{-n\beta(\epsilon_k - \mu)} = \frac{1}{1 - e^{-\beta(\epsilon_k - \mu)}}$
- Fermions ($n_k = 0$ or 1 by Pauli): $\Xi_k^F = 1 + e^{-\beta(\epsilon_k - \mu)}$

Substrate-mechanism reading: occupation number n_k corresponds to the substrate's K-type multiplicity. Bosons can multiply-occupy (substrate Pin(2) integer-weight K-types accept additive amplitudes); fermions cannot (substrate Pin(2) half-integer-weight K-types are excluded by antisymmetrization, Volume 5 Chapter 9).

7.2 Bose-Einstein distribution

For bosons, the mean occupation of mode k :

$$\langle n_k \rangle_{BE} = \frac{1}{e^{\beta(\epsilon_k - \mu)} - 1}$$

Bose-Einstein distribution. Always positive. Diverges as $\epsilon_k \rightarrow \mu$ — the signature of Bose-Einstein condensation.

7.3 Fermi-Dirac distribution

For fermions, the mean occupation:

$$\langle n_k \rangle_{FD} = \frac{1}{e^{\beta(\epsilon_k - \mu)} + 1}$$

Always between 0 and 1 (Pauli). At $T = 0$: step function — $n_k = 1$ for $\epsilon_k < \mu$, $n_k = 0$ for $\epsilon_k > \mu$. The Fermi energy $E_F = \mu(T = 0)$ separates filled from empty states.

7.4 Blackbody radiation

A blackbody is a perfect thermal-radiation emitter. The radiation field is a Bose gas (photons are bosons, spin 1) with $\mu = 0$ (photons can be freely created/destroyed). The Bose-Einstein distribution for photons gives:

$$\langle n(\omega) \rangle = \frac{1}{e^{\hbar\omega/k_B T} - 1}$$

Spectral energy density (Planck distribution):

$$u(\omega, T)d\omega = \frac{\hbar\omega^3}{\pi^2 c^3} \cdot \frac{d\omega}{e^{\hbar\omega/k_B T} - 1}$$

Limiting cases: - $\hbar\omega \ll k_B T$ (Rayleigh-Jeans): $u \approx \omega^2 k_B T / (\pi^2 c^3)$ — classical equipartition - $\hbar\omega \gg k_B T$ (Wien): $u \approx \hbar\omega^3 e^{-\hbar\omega/k_B T} / (\pi^2 c^3)$ — exponential cutoff

Integrated spectrum (Stefan-Boltzmann): $u_{\text{total}} = (8\pi^5/15)(k_B T)^4 / (hc)^3 = aT^4$ with a the radiation constant.

Substrate-mechanism reading: the substrate's photon K-types (spin 1, massless, Pin(2) integer-weight) are populated thermally per the Bose-Einstein distribution. The blackbody spectrum is the substrate's photon-K-type thermal distribution function.

7.5 The ultraviolet catastrophe and Planck's revolution

Classical Rayleigh-Jeans for blackbody predicted $u(\omega) \propto \omega^2 T$ — diverges as $\omega \rightarrow \infty$ (ultraviolet catastrophe). Real blackbodies have a high-frequency cutoff. Planck 1900 introduced the quantization hypothesis $E = n\hbar\omega$ to fix the divergence; this was the first appearance of $\hbar$ in physics.

Substrate-mechanism reading: the discrete substrate K-type spectrum (Volume 5 Chapter 4) intrinsically cuts off high-energy modes. The substrate doesn't have "infinite resolution" — its K-types are quantized; high- ω modes are exponentially suppressed (Wien limit). No ultraviolet catastrophe at the substrate level.

7.6 Bose-Einstein condensation

For a gas of N bosons in 3D below the critical temperature

$$T_c = \frac{1}{k_B} \left(\frac{n}{\zeta(3/2)} \right)^{2/3} \frac{h^2}{2\pi m}$$

(with $n = N/V$), a macroscopic fraction of particles condense into the ground state. The condensate fraction:

$$\frac{N_0}{N} = 1 - (T/T_c)^{3/2}$$

for $T < T_c$.

First observed experimentally Cornell-Wieman-Ketterle 1995 (Nobel 2001) — rubidium and sodium atomic gases at nanokelvin temperatures.

Substrate-mechanism reading: BEC is the substrate's macroscopic occupation of the ground-state K-type. Below T_c , the substrate's K-type Boltzmann distribution can no longer accommodate all N atoms in excited K-types; the remainder pile into the ground state. The condensate is the substrate's coherent K-type accumulation.

7.7 Fermi gases and white dwarfs

For non-relativistic fermions at $T = 0$: all states below the Fermi energy E_F are filled. Free-electron gas:

$$E_F = \frac{\hbar^2}{2m}(3\pi^2 n)^{2/3}$$

Mean energy per particle: $\langle E \rangle = (3/5)E_F$. Degeneracy pressure $P = (2/5)nE_F$.

Astrophysical application — white dwarfs: electron degeneracy pressure supports the star against gravity. Chandrasekhar mass limit $M_{Ch} \approx 1.4M_\odot$ — above this, electron degeneracy fails and the star collapses to neutron star or black hole.

Substrate reading: the Fermi gas's degeneracy is the substrate's Pin(2) half-integer-weight K-type filling pattern; the substrate cannot multiply-occupy fermion K-types, producing the degeneracy pressure.

7.8 Heat capacity of solids: Einstein and Debye

Classical equipartition gives $C_V = 3R$ per mole (Dulong-Petit) for solids — but experimentally $C_V \rightarrow 0$ as $T \rightarrow 0$. Einstein 1907: model atoms as quantum harmonic oscillators with frequency ω_E . Each oscillator contributes $\hbar\omega_E/(e^{\hbar\omega_E/k_B T} - 1)$ to internal energy. At low T : $C_V \propto e^{-\hbar\omega_E/k_B T}$ — exponentially small.

Debye 1912: extended to phonon modes with frequencies from 0 to a Debye cutoff ω_D . Gives $C_V \propto T^3$ at low T — matches experiment.

Substrate-mechanism reading: solids are substrate K-type lattices; phonons are substrate vibrational K-types (Volume 9 Chapter 7). Low- T heat capacity is the substrate's exponential suppression of high-energy K-type modes.

7.9 Worked example: photon gas in a cavity

For thermal radiation in a cubical cavity of side L at temperature T : - Photon density: $n_\gamma = (2\zeta(3)/\pi^2)(k_B T/\hbar c)^3$ - Energy density: $u = aT^4$ with $a = \pi^2 k_B^4/(15\hbar^3 c^3)$ - Entropy density: $s = (4/3)u/T$ - Pressure: $P = u/3$

For $T = 300$ K: $n_\gamma \approx 5 \times 10^{14}/\text{m}^3$ — about half a quadrillion photons per cubic meter at room temperature. Substrate reading: the substrate has 5×10^{14} photon-K-type quanta occupied per m^3 at thermal coupling 300 K.

7.10 Pauli paramagnetism and Landau diamagnetism

For an electron gas in a magnetic field B : - Spin Zeeman energy: $\pm\mu_B B$ - More electrons spin-aligned with $B \rightarrow$ paramagnetic susceptibility $\chi_P > 0$ - Orbital motion: cyclotron orbits in B ; spread over B direction gives diamagnetic susceptibility $\chi_D < 0$ - Net: $\chi_{\text{total}} = \chi_P - \chi_D > 0$ for most metals

Substrate reading: spin Zeeman is the substrate's Pin(2) coupling to external EM K-types (Volume 5 Chapter 3); cyclotron orbits are substrate Lorentz-coupled K-type motion.

7.11 K-audit anchors

- Volume 5 Chapter 9: spin-statistics theorem from substrate Pin(2) double cover
- Chapter 5 of this volume: partition function on D_{IV}^5
- Casey's Principle: counting at depth 0 $\rightarrow$ bosons count multi-occupancy, fermions count single-occupancy
- Mayer-Jensen 1949 (L1 ESTABLISHED): nuclear shell model (Fermi gas application)

Level 3 — 5th-grader accessibility

Quantum particles come in two flavors: bosons (photons, gluons, Higgs) and fermions (electrons, quarks, neutrinos). Bosons can pile up in one state (Bose-Einstein condensation: cool a gas of bosons enough and they ALL fall into the lowest state, becoming one big quantum object). Fermions can't share (Pauli exclusion: each state holds at most one fermion). This explains: - Blackbody radiation: photons (bosons) fill modes according to BE

distribution; that's the Planck curve, no UV catastrophe because high-frequency modes are quantum-suppressed. - Specific heats: at low T, the substrate can't reach excited K-type modes; heat capacities drop to zero (the third law). - White dwarfs: electrons (fermions) can't pile up; their degeneracy pressure holds up the star against gravity. Above 1.4 solar masses, gravity wins → neutron star or black hole. - Lasers: photons (bosons) like to share, so once you get a few photons in one mode, more pile in. That's stimulated emission, the basis of lasers.

All of this comes from the substrate's $Pin(2)$ double cover deciding which particles can share K-types.

What comes next

Chapter 8 develops phase transitions — order parameters, mean-field theory, Ising model, critical exponents.

Where to look this up

- Bose-Einstein, Fermi-Dirac: Pathria Ch 6-8; Huang Ch 8-9
- BEC: Cornell-Wieman-Ketterle 1995; Pitaevskii and Stringari, *Bose-Einstein Condensation*
- White dwarfs: Chandrasekhar 1931; Shapiro and Teukolsky, *Black Holes, White Dwarfs, and Neutron Stars*
- Debye model: Debye 1912; Ashcroft and Mermin Ch 23
- BST anchors: Volume 5 Chapter 9 spin-statistics; Chapter 5 partition function